The Duality of Culture Concepts: A Two-Mode Network Analysis

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Abstract

Kroeber and Kluckhohn's landmark 1952 monograph, *Culture: A Critical Review of Concepts and Definitions* (Kroeber and Kluckhohn, 1952), stands as the canonical mid-century effort to systematize the culture concept across anthropology and sociology. Beyond compiling over a hundred definitions, Kroeber and Kluckhohn indexed the substantive concepts comprising each definition and proposed an influential deductive taxonomy spanning seven broad categories. Rather than evaluating these definitions as isolated verbal pronouncements, I analyze the corpus as a two-mode network linking 171 definitions to 470 indexed concepts, exploiting the duality between definitions and concepts (Breiger, 1974). Maximum spanning trees of the dual projections reveal that the conceptual space is anchored by a steep core-periphery architecture, where an "ur-vocabulary" of foundational concepts (**social**, **society**, **group**, **behavior**) provides common currency across theoretical boundaries. Structurally central definitions, most notably by Gillin and Bidney, achieve high betweenness through synthetic integration across normative, psychological, and genetic registers, whereas historically revered formulations like Tylor's occupy a specialized peripheral position. Finally, unsupervised latent Dirichlet allocation reveals that Kroeber and Kluckhohn's deductive taxonomy fails to capture the latent thematic organization of the actual texts, operating instead as an external editorial sorting scheme rather than an organic structure of mid-century culture theorizing.

1 Introduction

In 1952, anthropologists Kroeber and Kluckhohn published their landmark monograph, *Culture: A Critical Review of Concepts and Definitions* (Kroeber and Kluckhohn, 1952), hereafter K&K. Appearing at a pivotal moment of postwar disciplinary expansion and theoretical consolidation, the volume sought to establish intellectual order across anthropology, sociology, and social psychology by systematically assembling, comparing, and classifying every extant scholarly definition of culture. To tame this sprawling archive, K&K devised a dual classification scheme. Inductively, they compiled an index of 470 constituent concepts or "sub-elements" (Index B) appearing across 171 definitions. Deductively, they sorted the definitions into seven overarching categories, spanning A Descriptive, B Historical, C Normative, D Psychological, E Structural, F Genetic, and G Incomplete.[1]

For decades, scholars have cited K&K's compilation as evidence either of hopeless conceptual fragmentation or of an emergent consensus across the human sciences. Yet assessments of the volume have remained almost entirely qualitative, impressionistic, and centered on isolated textual excerpts. In this paper, I adopt a relational perspective, treating K&K's archive not as a passive catalog but as a formal two-mode network connecting definitions to concepts (Breiger, 1974; Everett

[1] K&K further divide three of the seven categories into numbered subcategories, splitting Normative (C) into C-I and C-II, Psychological (D) into D-I through D-IV, and Genetic (F) into F-I through F-IV. The other four categories have no subcategories. Definitions that sit in a subcategory carry its numeral, as in C-I-18a or F-II-4 below.

and Borgatti, 2013). The relational formulation rests on the principle of duality, where definitions are constituted by the specific configurations of concepts they mobilize, while concepts derive their disciplinary meaning and structural significance from the company of definitions in which they co-occur.

Treating the archive as a two-mode network allows an investigation of several fundamental questions in the cognitive history of social scientific concepts. First, what is the macro-topology of mid-century culture theorizing? Does the conceptual field exhibit a fragmented, modular structure composed of insular theoretical camps, or does it coalesce around a shared core-periphery architecture? Second, which specific definitions and concepts serve as structural bridges binding divergent intellectual traditions together? Third, does K&K's deductive taxonomy reflect genuine latent boundaries in the actual wording of the definitions, or was their classification an artificial editorial schema imposed from above?

To answer these questions, I investigate both the structural backbone of the network using maximum spanning trees across dual one-mode projections and the latent semantic structure of the definition texts using latent Dirichlet allocation (LDA). The definitions network links 470 concepts to 171 definitions, with 169 definitions drawn from K&K's primary author index and two (C-I-18a and D-II-16) identified from main-text citations. Following Boroch (2016), four composite definitions are split into distinct quotations, yielding the 171 analyzed definitions.[2] The full network contains 10 connected components, with the largest component containing 626 nodes (166 definitions and 460 concepts) and 1195 of the 1201 edges. The analyses that follow focus on the 2-core of this largest component, comprising 149 definitions and 160 concepts connected by 873 edges, which preserves all mutually reinforcing structural pathways while pruning isolated and pendant peripheral nodes.

2 Macro-Topology: The Core-Periphery Architecture of Culture Concepts

Table 1 reports global summary statistics for the definitions network, and Figure 1 displays the degree distributions for both modes across the full network. The network displays a relatively low density of 0.015, indicating that only 1.5% of all potential connections between concepts and definitions are realized. The overall sparsity, however, conceals a marked structural asymmetry between the two modes, characterized by an extreme divergence in their connectivity profiles.

The concept degree distribution, plotted on the left panel of Figure 1, exhibits the classic heavy-tailed signature of a core-periphery structure. The vast majority of concepts appear in only a handful of definitions. Fully 79% of all indexed concepts occur in two or fewer definitions, and the median concept appears in only a single definition. At the opposite extreme sits a compact, highly prominent cluster of "super-connector" concepts that permeate mid-century culture theorizing. As documented in Table 2, concepts like **social** (degree 35 in the 2-core), **society** (31), **group** (27), **behavior** (24), **total** (22), **ideas** (21), **beliefs** (19), and **customs** (16) serve as ubiquitous building blocks across theoretical traditions.[3] The heavy-tailed structure shows that mid-century culture theorists did not generate novel conceptual vocabularies from scratch; rather, they anchored their

[2] Details regarding the transcription of Index B, OCR reconciliation, the variant splits, and edge resolution are documented in Appendix Appendix A.

[3] Over the full range of concept degrees, a discrete power law fits significantly better than a truncated exponential (Vuong $z = 6.1$, $p < 0.001$; $\hat{\alpha} = 2.13$), suggesting a mechanism analogous to preferential attachment in scientific vocabulary accumulation (Barabási and Albert, 1999). The advantage attenuates for degrees of 2 and above ($z = 1.9$, $p = 0.06$) and vanishes for degrees of 3 and above ($|z| < 1$), indicating that the heavy tail reflects a discrete tier of foundational vocabulary rather than an unconstrained scale-free process across the entire lexicon.

Table 1: Summary statistics for the definitions network. Statistics describe the full network, before the restriction to the largest component's 2-core used in the analyses. Average nearest-neighbor degree (k_{nn}) is the average degree of a node's neighbors; degree correlation shows how a node's degree relates to the degree of its neighbors. Degree variances are sample variances (divisor $n-1$) over all nodes of each type, including isolates.

	Network
Edges	1201
Density	0.015
Degree correlation r	−0.076
Concepts ($N = 470$)	
Mean degree	2.56
Minimum degree	1
Maximum degree	36
Degree variance	16.34
Mean k_{nn}	10.18
Definitions ($N = 171$ definitions, 5 isolated)	
Mean degree	7.02
Minimum degree	0
Maximum degree	25
Degree variance	20.52
Mean k_{nn}	9.22

formulations within a deeply entrenched, shared "ur-vocabulary" centered on collective life, social organization, and regular patterns of behavior.

In contrast to the skewed concept distribution, the definition degree distribution (right panel of Figure 1) exhibits a symmetric, approximately bell-shaped profile centered around a mean of 7.02 concepts per definition (or 5.9 on the 2-core; median 5). This indicates that mid-century scholars adhered to a consistent norm of conceptual economy, where definitions were rarely minimalist single-concept formulas, nor were they boundless, undisciplined catalogs. Instead, the typical definition assembled between four and eight concepts, with the most expansive formulations topping out at 14 or 15 concepts (e.g., Reuter 1939; Taylor 1948; Hiller 1933; Gillin and Gillin 1942).[4]

The degree correlation between the two modes is slightly negative ($r = -0.076$), indicating a subtle disassortative tendency where peripheral concepts tend to attach to broad, multi-concept definitions, while peripheral definitions lean heavily on ubiquitous core concepts.[5] The relational dynamic is further illuminated by the nearest-neighbor degree statistics (k_{nn}). The average neighboring definition of an indexed concept contains 10.18 concepts, while the average neighboring concept of a definition appears in 9.22 definitions. Rather than splintering into isolated intellec-

[4] Across all 171 definitions, quotation length correlates moderately with 2-core concept degree (Pearson $r = 0.55$, Spearman $\rho = 0.57$; $r = 0.60$ and $\rho = 0.61$ among the 149 definitions in the 2-core). When calibrated by quotation length, as shown in the rightmost columns of Table 2, definitions by Dixon 1928 (A-3(a)), Kluckhohn 1942 (D-II-4), Young 1934 (D-III-2), and Bain 1942 (F-III-1) emerge as remarkably dense, packing over four distinct concepts into every ten words.

[5] Against a degree-preserving null model that randomizes bipartite affiliations while preserving both marginal degree sequences (1000 draws; seed 6128), the expected correlation is −0.033 with standard deviation 0.022 ($z = -1.94$, two-sided $p = 0.041$). The observed negative correlation is slightly stronger than degree constraints alone would dictate.

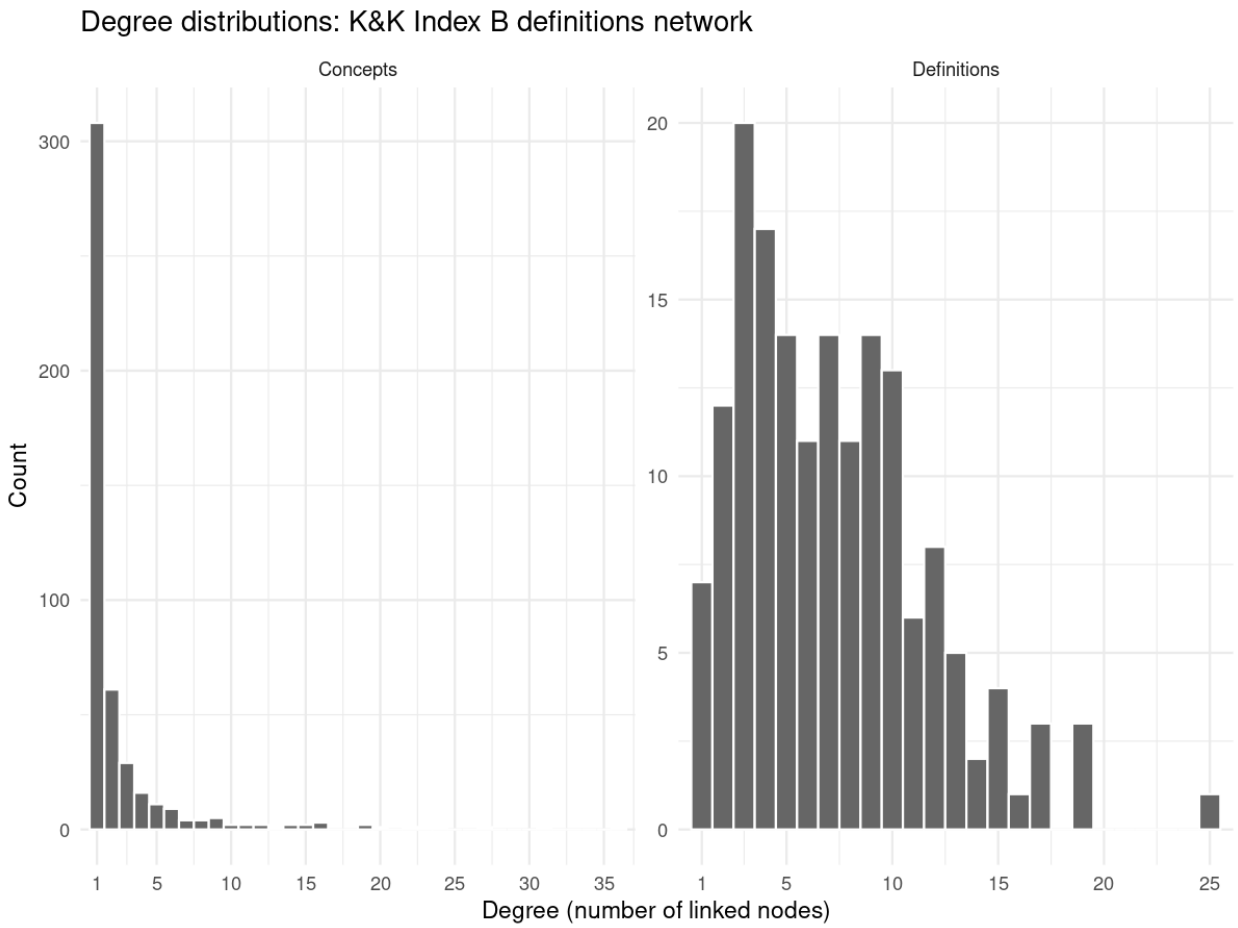


Figure 1: Degree distributions of the definitions network. On the left, sub-element (concept) nodes; on the right, definition nodes.

tual cliques, mid-century culture theorizing achieved macro-structural integration through a dual relational mechanism, where definitions gained breadth by linking to widely shared concepts, while concepts gained relevance by anchoring comprehensive definitions.

3 Structural Backbones: Spanning Trees of the Dual Projections

Because the bipartite affiliation matrix contains no direct ties between pairs of concepts or pairs of definitions, I examine the latent connections within each mode through dual one-mode projections (Breiger, 1974; Everett and Borgatti, 2013). Following standard practice, two concepts are tied if they co-occur within at least one definition, with the edge weight reflecting the number of shared definitions; dually, two definitions are tied through the shared concepts comprising them. In matrix notation, these dual projections are given by

$$W_{\text{concept}} = AA^\top, \qquad W_{\text{def}} = A^\top A, \tag{1}$$

where $(W_{\text{concept}})_{ij} = \sum_k a_{ik}a_{jk}$ represents the co-occurrence weight between concepts i and j, and $(W_{\text{def}})_{kl} = \sum_i a_{ik}a_{il}$ captures the conceptual overlap between definitions k and l. Both projections are derived from the connected 2-core of the largest component (160 concepts and 149 definitions).

Table 2: Highest-degree nodes of the definitions network on the LCC 2-core. Concept degree counts the 2-core definitions that contain the concept; definition degree counts the 2-core concepts the definition contains. The last two columns rank definitions by degree per 10 words of quotation, using the English translation where one exists. Per-word rankings favor short quotations; across the 171 definitions the per-word value correlates −0.23 with word count. Ties at the cutoff are included.

Concepts, by degree		Definitions, by degree		Definitions, by degree per 10 words	
Concept	Degree	Definition	Degree	Definition	Per 10 words
SOCIAL	35	F-I-8 (Reuter 1939)	15	A-3(a) (Dixon 1928)	5.00
SOCIETY	31	F-II-8 (Taylor 1948)	15	D-II-4 (Kluckhohn 1942)	4.29
GROUP	27	A-8 (Hiller 1933)	14	D-III-2 (Young 1934)	4.29
BEHAVIOR	24	C-I-6 (Gillin and Gillin 1942)	14	F-III-1 (Bain 1942)	4.29
TOTAL	22	C-II-4 (Bidney 1946)	13	A-1 (Tylor 1871)	4.14
IDEAS	21	D-I-7 (Panunzio 1939)	13	B-13 (Groves and Moore 1940)	3.75
BELIEFS	19	E-3 (Ogburn and Nimkoff 1940)	13	G-4 (Rouse 1939)	3.75
INDIVIDUALS	19	F-I-19 (Murdock 1949)	13	F-II-6 (Kluckhohn and Kelly 1945)	3.64
SUM	19	F-I-3 (Folsom 1931)	13	B-6 (Malinowski 1931)	3.46
CUSTOMS	16	A-1 (Tylor 1871)	12	A-3(b) (Dixon 1928)	3.33
LEARNING	16	C-I-17 (Frank 1948)	12	F-I-10 (Dodd 1941)	3.33
				F-I-15 (Carr 1945)	3.33

Note: Degrees are computed on the LCC 2-core used in the analyses (149 definitions and 160 concepts). Table 1 and Figure 1 describe the full network instead. The 22 definitions outside the 2-core have degree 0 and a per-10-word value of 0. Per-10-word values are definition degree × 10 divided by the word count of the quotation. Ties at the cutoff are included, so the three lists have different lengths. F-I-10 is Dodd 1941 as Boroch attributes it, a 1941 committee mimeograph of four authors.

Dual one-mode projections are notoriously dense and redundant, making it difficult to discern their primary structural pathways (Neal, 2014). To extract the underlying structural backbone of each projection, I compute its maximum spanning tree (MST). An MST identifies the minimally connected subgraph spanning all nodes that maximizes the total sum of retained edge weights ($\sum_{e \in T} w_e$), effectively filtering out dense cycles while preserving the strongest structural bridges (Kruskal, 1956). Because tie weights frequently recur in discrete projection data, the maximum spanning tree is not mathematically unique. To guard against artifacts of arbitrary tie-breaking, I construct an ensemble of 1000 random maximum spanning trees for each projection, resolving weight ties in a randomized sequence using Kruskal's algorithm (Kruskal, 1956). For each node, I compute its mean Freeman-normalized betweenness centrality across the 1000 trees (Freeman, 1978), along with the percentage of trees in which the node ranks in the top ten (the top-10 share).

Table 3 displays the ten most central concepts and definitions ranked by ensemble mean betweenness. Figure 2 and Figure 3 display the representative tree member closest to the ensemble mean for concepts and definitions, respectively, with node sizes proportional to ensemble betweenness and colors indicating modular communities detected via the leading-eigenvector method (Newman, 2006).

3.1 The Semantic Geography of the Concept Tree

The concept backbone (Figure 2) reveals how 160 core concepts arrange themselves across the semantic landscape. Rather than a diffuse, undifferentiated web, the concept tree is structured around a small set of high-betweenness trunk nodes that mediate between distinct semantic regions. Across the ensemble, 21 concepts serve as terminal leaves in every single tree, and the median

Table 3: Top 10 nodes by mean Freeman-normalized betweenness over 1000 random maximum spanning trees of each projection. Degree is the node's degree in the 2-core. Top-10 share is the percentage of trees in which the node ranks in the top ten. Concepts and definitions are ranked separately. Ties are broken at random in each tree, so the ranks below the first place are sensitive to tie-breaking. A-10 (10th, share 45%) and A-1 (11th, share 50%) differ by 1.1 points in mean betweenness.

Node type	Node	Author-year	Mean norm. betw. (%)	Top-10 share (%)	Degree
Concepts	GROUP		66.0	100	27
	SOCIAL		57.8	100	35
	SOCIETY		54.7	100	31
	BEHAVIOR		51.6	100	24
	CUSTOMS		36.9	100	16
	BELIEFS		25.7	97	19
	MEMBERS OF A SOCIETY		21.5	99	14
	HABITS		19.3	92	15
	IDEAS		15.8	60	21
	LEARNING		12.5	51	16
Definitions	C-I-6	Gillin and Gillin 1942	77.2	100	14
	A-16	Bidney 1947	35.3	77	11
	F-I-3	Folsom 1931	33.3	86	13
	C-II-4	Bidney 1946	29.1	77	13
	F-I-19	Murdock 1949	28.5	71	13
	C-II-3	Bidney 1942	26.9	71	10
	F-II-8	Taylor 1948	25.9	52	15
	A-8	Hiller 1933	24.1	77	14
	C-I-4	Klineberg 1935	21.8	70	8
	A-10	Linton 1936	21.3	45	11

Note: Each definition has an identifier in the format X-Y-Z, where X is a letter (A–G) for K&K's seven definition categories (Descriptive, Historical, Normative, Psychological, Structural, Genetic, Incomplete), Y is a roman numeral for the subcategory, and Z is the number of the definition within it. Variants such as A-3(a) and A-3(b) are the two quotations of one K&K definition, and they are counted as separate definitions (see Appendix Appendix A). The author–year column names the source of each definition's quotation, as Boroch attributes it; the same author and year label the node in Figure 3.

concept acts as a leaf in 76% of trees. High degree is positively associated with tree betweenness (Spearman $\rho = 0.71$), yet degree alone does not determine structural centrality. Several concepts with 15 or more connections exhibit negligible betweenness. Specifically, *total* (degree 22) achieves a mean betweenness of only 7.3%, *sum* (19) achieves 2.6%, and *transmission* (15) achieves a mere 0.9%. By contrast, **ideas** (degree 21) registers 15.8% betweenness, and **customs** (degree 16) commands 36.9%, ranking fifth overall. While *total* and *sum* serve as generic syntactic qualifiers across many definitions, concepts like **customs** and **ideas** function as crucial semantic bridges linking distinct thematic domains.

Substantively, the concept backbone organizes mid-century culture discourse into four primary regional domains.

The first domain assembles the structural and systemic vocabulary, anchored by **group** (66.0% betweenness), which forms the hub of an expansive community encompassing *communication*, *organization*, *folkways*, *systems*, *designs for living*, and *reactions*. The regional vocabulary reflects the structuralist and configurationalist emphasis on culture as patterned social organization.

The second domain carries the institutional and heritage vocabulary, anchored by **social** (57.8% betweenness), which coordinates a cluster including *social heritage*, *institutions*, *social needs*, *pat-*

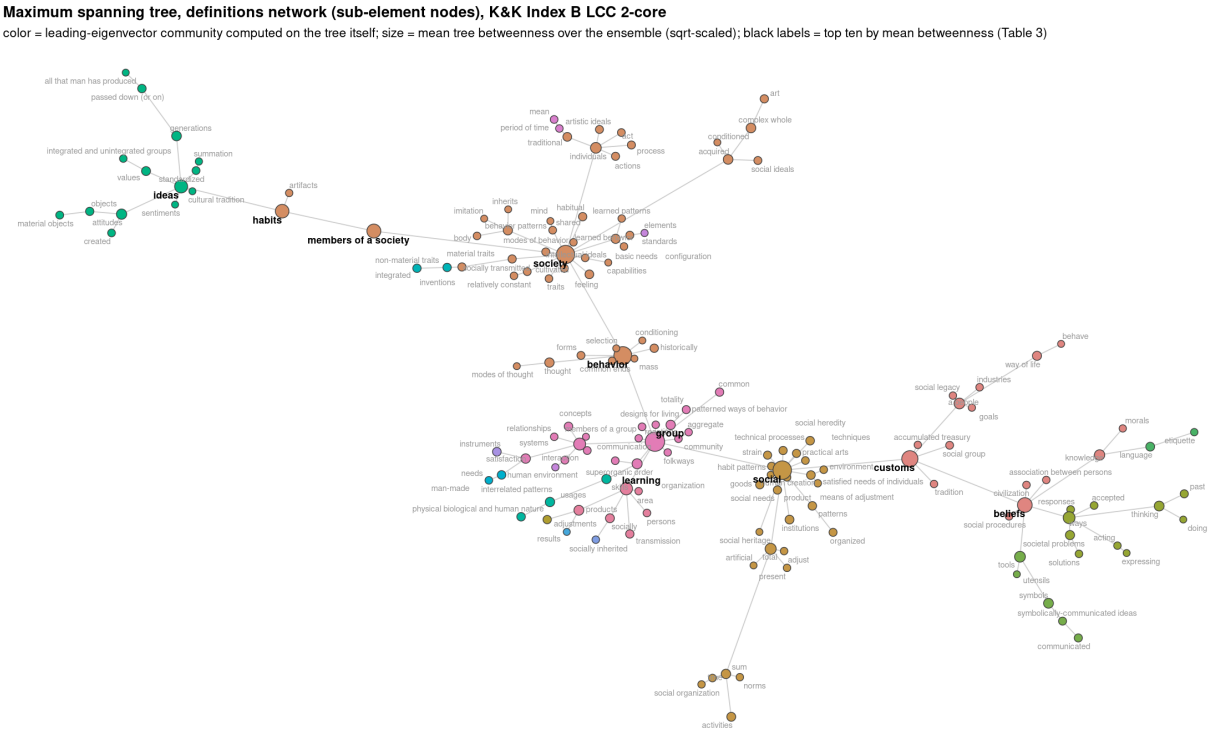


Figure 2: Maximum spanning tree of the definitions network, sub-element (concept) nodes. Color is the leading-eigenvector community on the tree shown; size is mean betweenness over the tree ensemble.

terns, and *human creation*. The region connects directly to **customs** (36.9%), which heads a smaller branch containing *accumulated treasury*, *social group*, and *tradition*.

The third domain gathers the behavioral and psychological vocabulary, anchored jointly by **society** (54.7% betweenness) and **behavior** (51.6% betweenness). The **society** hub organizes concepts referring to socialized actors (*individuals*, *shared*, *basic needs*), while **behavior** links to cognitive and intentional dimensions (*thought*, *forms*, *historically*, *selection*, *common ends*).

The fourth domain holds the normative and ideational vocabulary, mediated by **beliefs** (25.7% betweenness), which links *knowledge*, *civilization*, and *social procedures* to the rest of the tree, and by **ideas** (15.8%), which heads a branch encompassing *attitudes*, *sentiments*, and *values*. The intermediate concepts **habits** (19.3%) and **members of a society** (21.5%) serve as pivotal structural joints connecting ideational vocabulary to the behavioral **society** hub.

Notably, concepts related to cultural acquisition and transmission occupy a surprisingly peripheral branch of the backbone. Concepts such as **learning** (12.5% betweenness), *transmission*, *socially*, *products*, and *persons* do not form an independent central axis; rather, they hang as a subordinate appendage off the **group** hub. This topological arrangement reveals an important feature of mid-century thinking about the culture concept, where theorizing focused primarily on characterizing culture as an established social-structural and behavioral entity, treating mechanisms of individual learning and transmission as secondary, derivative processes.In that respect recent thinking in culture and cognition studies in sociology have made central this previously peripheral emphasis (Lizardo, 2017).

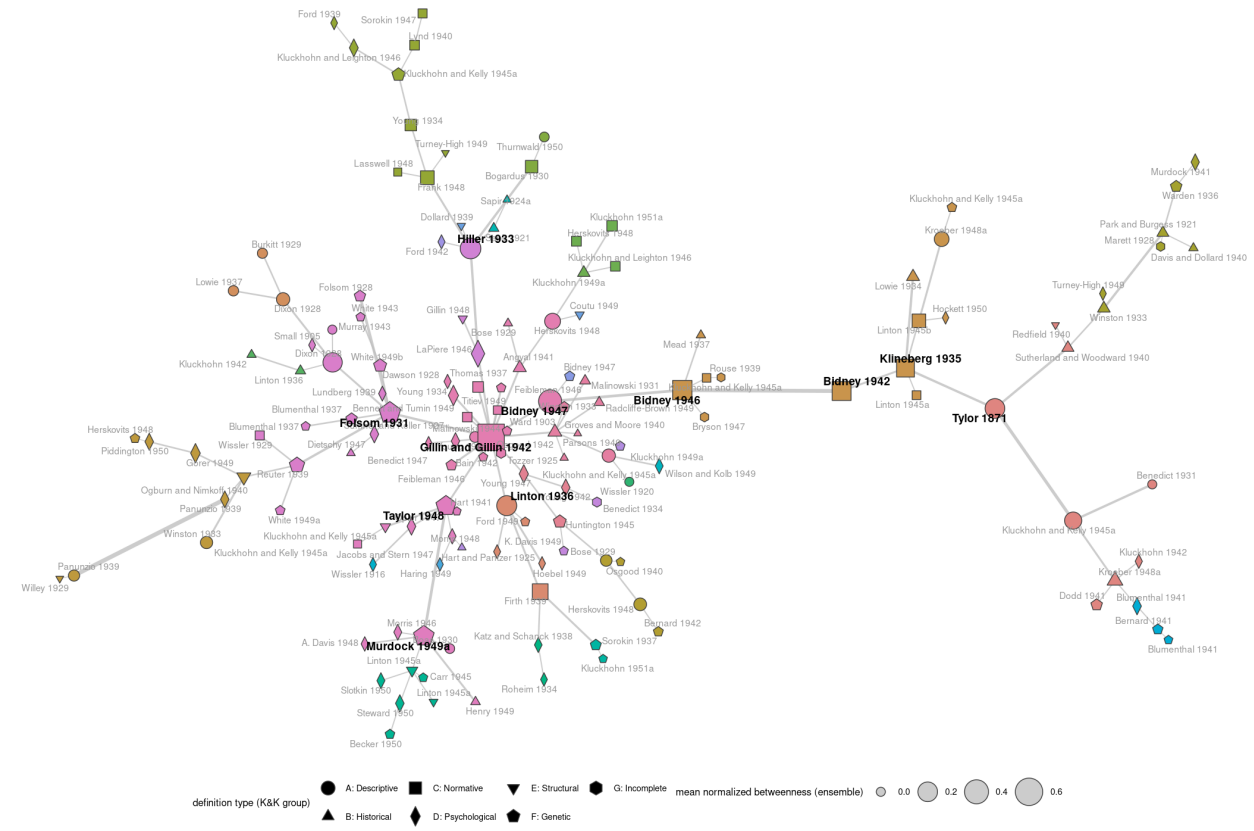


Figure 3: Maximum spanning tree of the definitions network, definition nodes labeled by the author and year of their quotation. Shape is the K&K definition group (A–G, Descriptive to Incomplete); color is the leading-eigenvector community on the tree shown; size is mean betweenness over the tree ensemble.

3.2 Bridging Definitions: Synthetic Theorizing vs. Historical Origin

The definition-level tree (Figure 3) displays the structural backbone connecting 149 definitions. High betweenness in this projection highlights definitions that act as theoretical bridges, binding together disparate conceptual vocabularies across the corpus. Seven of the top ten definitions in Table 3 belong to K&K's Normative (C) or Genetic (F) categories, while the remaining three (A-16, A-8, A-10) are Descriptive.

The single most prominent structural bridge in the entire corpus is Gillin's 1942 definition (C-I-6), commanding a remarkable mean betweenness of 77.2% and appearing in the top ten across 100% of the tree ensemble. Gillin's quotation sits at the center of the largest community in Figure 3, linking definitions from six of K&K's seven categories. Crucially, Gillin's preeminent bridging status is not a trivial consequence of text length. At 64 words, the quotation sits at the 89th percentile of length, well below the longest definitions in the corpus (which reach 142 words). Rather, Gillin's centrality reflects a deliberate conceptual synthesis, weaving together customs, traditions, attitudes, ideas, symbols, social behavior, group patterns, and generational transmission into a single cohesive formulation. Accordingly, Gillin defines culture as follows:

> The **customs**, **traditions**, **attitudes**, **ideas**, and **symbols** which **govern social**

> **behavior** show a wide variety. Each **group**, each **society** has a set of **behavior patterns** (overt and covert) which are more or less **common** to the members, which are passed down from **generation** to **generation**, and **taught** to the **children**, and which are constantly liable to **change**. These **common** patterns we call the culture... (Gillin and Gillin 1942)

Gillin thus hits all the right notes, talking about the contents of culture (customs, traditions, attitudes, ideas, symbols), the social and behavioral processes through which culture is enacted (governing social behavior, common to members), and the generational mechanisms through which culture is transmitted (passed down, taught to children). Thus, while 93 other definitions share at least one of Gillin's core concepts, their average betweenness is only 4.7%. Gillin's formulation achieves maximum structural prominence because it operates as a grand intersection, coordinating the normative, psychological, structural, and genetic vocabularies of mid-century social science.

An even more notable finding is the theoretical centrality of philosopher and anthropologist David Bidney (1968). Bidney accounts for three of the top six bridging definitions in the corpus, namely A-16 (ranking 2nd with 35.3% betweenness), C-II-4 (4th with 29.1%), and C-II-3 (6th with 26.9%). Unlike sprawling descriptive inventories, Bidney achieves exceptional betweenness through concise, highly integrated formulations. Bidney's A-16 definition (Bidney 1947) is only 34 words long, yet it commands the second-highest betweenness in the network.

> [...] functionally and secondarily, culture refers to the **acquired** forms of technique, **behavior**, **feeling** and **thought** of **individuals** within **society** and to the **social institutions** in which they **cooperate** for the attainment of **common ends**. (Bidney 1947)

This definition also hits the big notes more succinctly than Gillin, talking about acquisition, the main elements of culture (behavior, feeling, thought), the social context (individuals, society, social institutions), and more notably, the functional *purpose* of culture (cooperation for common ends). Removing Bidney's A-16 definition from the representative tree shatters the network backbone into three disconnected components of 116, 31, and 1 definition. Bidney's other two central definitions, C-II-4 (40 words; 29.1% betweenness) and C-II-3 (40 words; 26.9% betweenness), share a distinct community that explicitly synthesizes psychological and normative registers.

> An integral or holistic concept of culture comprises the **acquired** or **cultivated behavior**, **feeling**, and **thought** of **individuals** within a **society** as well as the **patterns** or **forms** of intellectual, social, and artistic ideals which human societies have professed **historically**. (Bidney 1946)

> A culture consists of the **acquired** or **cultivated behavior** and **thought** of **individuals** within a **society**, as well as of the intellectual, artistic, and social ideals which the members of the society profess and to which they strive to conform. (Bidney 1942)

Bidney's structural centrality reveals a key dimension of mid-century concept formation around culture. In standard disciplinary histories, Bidney is often remembered as an eccentric philosophical critic of Boasian culture theory. Yet within the relational topology of the conceptual network, Bidney emerged as the preeminent synthetic theorist of the era. His self-conscious quest for an "integral or holistic" concept of culture led him to craft definitions that systematically reconciled individual psychological processes (acquired behavior, feeling, thought) with collective social facts

(institutions, ideals, historical patterns) and more "normative" conceptions of the values and goals instilled by the enculturation process.

Processual and genetic definitions also occupy prominent bridging positions. Folsom's 1931 definition (F-I-3; 75 words) ranks 3rd with 33.3% betweenness. It is distinctive in its attempt to bridge between material artifacts and social ideals, emphasizing the totality of human production and the generational transmission of culture:

> Culture is not any part of man or his inborn equipment. It is the sum total of **all that man has produced**: **tools**, **symbols**, most organizations, **common activities**, **attitudes**, and **beliefs**. It includes both **physical products** and **immaterial products**. It is everything of a **relatively permanent** character that we call **artificial**, everything which is **passed down** from one generation to the next rather than **acquired** by each generation for itself: it is, in short, **civilization**. (Folsom 1931)

Similarly, Murdock's 1949 formulation (F-I-19; 70 words) ranks 5th with 28.5% betweenness, explicitly framing culture as the synthetic outcome of the "interaction of learning and society":

> The **interaction** of **learning** and **society** thus produces in every human **group** a **body** of **socially transmitted adaptive behavior** which appears super–individual because it is **shared**, because it is **perpetuated** beyond the individual life span, and because its quantity and quality so vastly exceeds the capacity of any single person to achieve by his own unaided effort. The term culture is applied to such **systems** of **acquired** and transmitted **behavior**. (Murdock 1949)

In contrast to these central synthesizers, other prominent definitions occupy surprisingly peripheral structural positions. Kroeber's own definition (A-17; 65 words) ranks 25th in tree betweenness (5.7%), functioning as a local descriptive summary rather than an integrative global bridge.

> [. . .] the **mass** of learned and transmitted **motor reactions**, **habits**, **techniques**, **ideas**, and **values** – and the **behavior** they induce – is what constitutes culture. Culture is the special and exclusive product of men, and is their distinctive quality in the cosmos [. . .]. Culture [. . .] is at one and the same time the **totality** of products of social men, and a tremendous **force** affecting all human beings, **socially** and **individually**. (Kroeber 1948)

Even more instructive is the structural position of Tylor's foundational 1871 definition (A-1), widely canonized as the historical origin of the anthropological concept of culture (and thus placed first in K&K's Index B).

> Culture, or **civilization**, [. . .] is that **complex whole** which includes **knowledge**, belief, **art**, law, **morals**, custom, and any other **capabilities** and **habits acquired** by man as a member of **society**. (Tylor 1871)

Tylor's definition links to 12 concepts in the 2-core (**society**, **customs**, **beliefs**, **members of a society**, **habits**), yet it achieves a mean betweenness of only 20.2%, ranking 11th overall. In Figure 3, Tylor sits within an eight-definition cluster on the periphery of the network backbone, largely decoupled from the central bridges occupied by Gillin and Bidney. The peripheral position reveals a critical divergence between historical veneration and structural integration; while Tylor's formula provided the discipline of anthropology with its founding formulation, twentieth-century culture theorizing moved via synthetic bridges that integrated behavioral, psychological, and normative registers, leaving Tylor's classic definition as an honored but specialized peripheral outpost.

4 Data Driven Classification of Definitions

While K&K's Index B provides an inductive mapping of constituent concepts, K&K's primary organizational intervention was deductive, dividing definitions into seven broad categories (A Descriptive, B Historical, C Normative, D Psychological, E Structural, F Genetic, and G Incomplete). Did these categories capture genuine latent semantic boundaries in how scholars formulated definitions, or were they an external taxonomic grid imposed upon the texts?

To address this question without relying on K&K's manual index or predefined categories, I turn to latent Dirichlet allocation (LDA) (Blei et al., 2003). LDA models each text as an unsupervised mixture over latent topics, where each topic is characterized by a multinomial distribution over terms. To respect the multi-clause nature of complex definitions, I segment each quotation at sentence and semicolon boundaries and treat the individual clause as the unit of topic estimation. The definition itself remains the unit of analysis, since clause-level topic memberships are aggregated back to definitions before any comparison with K&K's categories. Filtering out clauses with fewer than three words, removing stopwords, grammatical particles, and high-frequency frame words (*culture*, *society*, *social*, *man*), and collapsing plurals to their singular forms produced a final corpus of 265 clauses across 171 definitions and a vocabulary of 340 terms. Stopwords, grammatical particles, and high-frequency frame words (*culture*, *society*, *social*, *man*) were removed, and plurals were collapsed to their singular forms. Retaining terms appearing in at least two clauses yielded a final corpus of 265 clauses across 171 definitions and a vocabulary of 340 terms.[6]

Following standard validation procedures, I fit an LDA model with $k = 12$ topics using Gibbs sampling, setting a sparse Dirichlet document-topic prior of $\alpha = 0.1$ to encourage crisp topic assignments (Griffiths and Steyvers, 2004).[7] To account for sampling variability, I estimate 50 independent runs, aligning topics across runs to a common reference fit via Hungarian matching on topic-term cosine similarities (achieving an average alignment similarity of 0.69). Each definition's final topic weights represent the clause-length-weighted average across all 50 aligned runs.

Table 4 lists the twelve estimated topics along with their leading terms and dominant definition counts, and Figure 4 displays the top ten terms for each topic. The topics capture distinct, substantively coherent semantic registers. Four major topics dominate the corpus, accounting for 83 of the 171 definitions, led by *transmitted group behavior* (26 definitions, led by *behavior*, *group*, *generation*), *way of life of a people* (20 definitions, led by *way*, *life*, *group*), *customs, beliefs and knowledge* (19 definitions, led by *custom*, *belief*, *people*), and *material tools and ideas* (18 definitions, led by *material*, *idea*, *includes*, *tool*). Smaller topics isolate more specialized theoretical idioms, such as *biological needs and nature* (9 definitions) and *cultural order of phenomena* (9 definitions).

4.1 Representative Definitions of the Latent Topics

Table 5 presents the five most representative definitions for each topic, ranked by their topic membership weight, along with their primary source works.

The highest-ranking definitions for each topic illustrate the substantive clarity of the model's semantic extraction.

Individual acquired behavior finds its clearest exemplar in C-II-3 (Bidney 1942; weight 0.51). "A culture consists of the **acquired** or cultivated **behavior** and thought of **individuals** within a

[6] One definition, C-I-7 (Simmons 1942), contained no term surviving the frequency filter and was assigned a uniform topic distribution.

[7] Held-out perplexity evaluated across $k = 8$ to 30 remains essentially flat, indicating that $k = 12$ provides an optimal balance between granular semantic interpretability and model parsimony.

Table 4: The twelve LDA topics, showing the assigned label for each, its five highest-weight ensemble terms, and the number of definitions whose dominant topic it is (the topic with the largest weight).

Topic	Label	Top terms	Definitions
1	individual acquired behavior	individual, behavior, pattern, acquired, habit	13
2	human products and results	human, product, result, activities, learned	12
3	transmitted group behavior	behavior, group, generation, learned, pattern	26
4	shared group living	group, common, men, relationship, action	9
5	biological needs and nature	mean, human, whole, biological, concept	9
6	acquired habits as complex wholes	habit, acquired, member, includes, individual	8
7	cultural order of phenomena	human, cultural, phenomena, order, animal	9
8	sum total of group life	sum, total, individual, mean, past	18
9	material tools and ideas	material, idea, includes, attitudes, tool	18
10	customs, beliefs and knowledge	custom, belief, people, tradition, knowledge	19
11	way of life of a people	way, life, group, people, things	20
12	man-made environment	behavior, environment, part, material, human	10

society, as well as of the intellectual, artistic, and social ideals which the **members** of the society profess and to which they strive to conform."

Human products and results receives its most concentrated expression in D-I-9 (Committee on Conceptual Integration 1941; weight 0.74). "Culture consists of all **results** (**products**) of **human learned effort** at **adjustment**."

Transmitted group behavior is headed by D-II-6 (Benedict 1947; weight 0.67). "[...] culture is the sociological term for **learned behavior**, **behavior** which in man is not given at birth, which is not determined by his germ cells as is the **behavior** of wasps or the social ants, but must be **learned** anew from grown people by each new **generation**."

Shared group living is best captured by B-5 (Bose 1929; weight 0.36). "[...] we may describe culture as including such behaviour as is **common** among a **group** of **men** and which is capable of transmission from **generation** to **generation** or from one country to another."

Biological needs and nature carries its highest weight on A-11 (Panunzio 1939; weight 0.86). "It [culture] is the complex **whole** of the **system** of **concepts** and usages, organizations, skills, and instruments by means of which mankind deals with **physical**, **biological**, and **human nature** in satisfaction of its **needs**."

Acquired habits as complex wholes is anchored by A-4 (Benedict 1929; weight 0.44). "[...] the **complex** whole which **includes** all the **habits acquired** by man as a **member** of society."

Cultural order of phenomena is dominated by F-IV-2 (Ostwald 1915; weight 0.67). "These

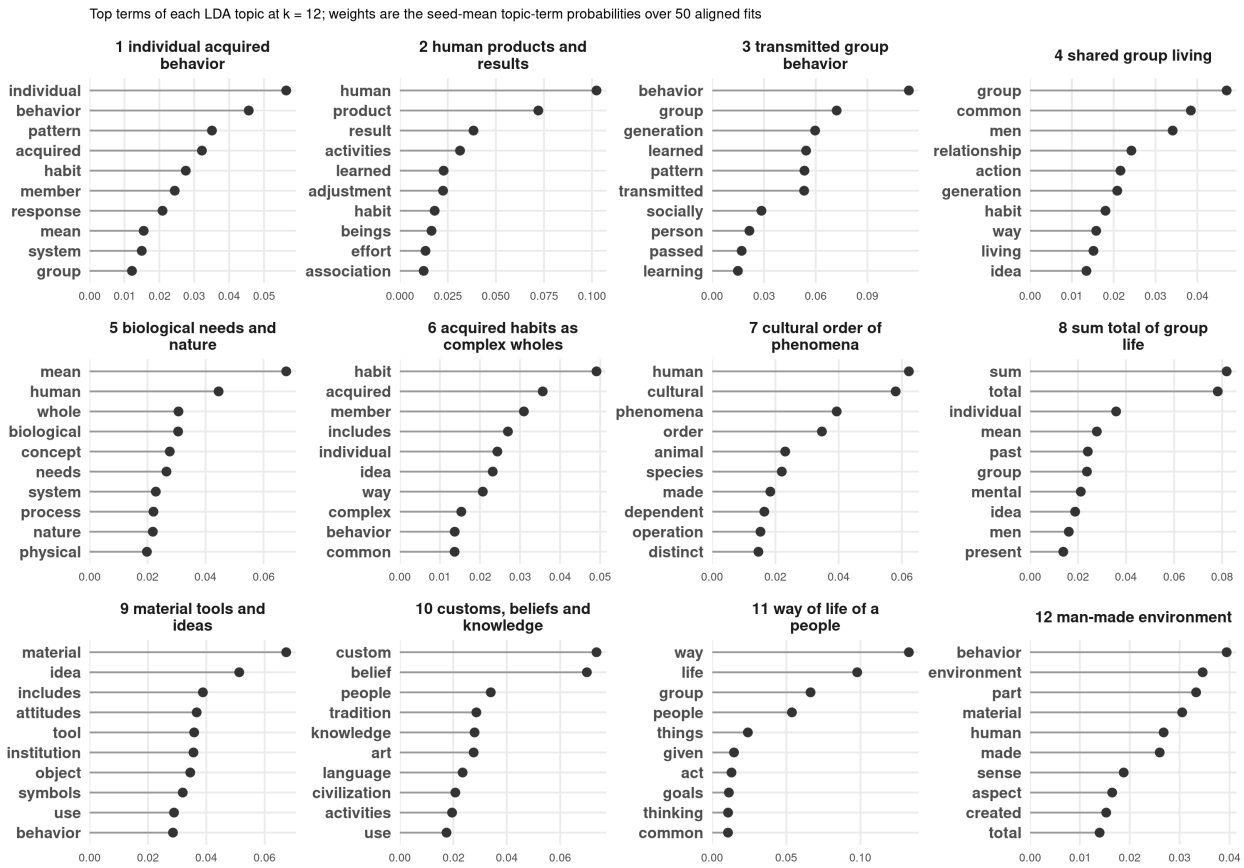


Figure 4: Top terms of each LDA topic at $k = 12$, labeled as in Table 4. Bars show the seed-mean topic-term probabilities over 50 aligned fits.

specifically **human** peculiarities which differentiate the race of the Homo sapiens from all other **species** of **animals** is comprehended in the name culture [. . .]."

Sum total of group life finds its leading formulation in A-10a (Lowie 1937; weight 0.48). "By culture we understand the **sum total** of what an **individual** acquires from his society—those beliefs, customs, artistic norms, food habits, and crafts which come to him not by his own creative activity but as a legacy from the **past**, conveyed by formal or informal education."

Material tools and ideas is exemplified by C-I-18 (Titiev 1949; weight 0.64). "[. . .] the term **includes** those **objects** or **tools**, **attitudes**, and forms of **behavior** whose **use** is sanctioned under given conditions by the members of a particular society."

Customs, beliefs and knowledge draws its highest weights from A-3(a) (Dixon 1928; weight 0.64). "The sum of all [a **people**'s] **activities**, **customs**, and **beliefs**."

Way of life of a people is epitomized by C-I-13 (Kluckhohn and Leighton 1946; weight 0.70). "A culture is any given **people**'s **way** of **life**, as distinct from the **life**-ways of other peoples."

Man-made environment is conveyed by F-I-1a (Willey 1927; weight 0.58). "[. . .] that **part** of the **environment** which man has himself **created** and to which he must adjust himself."

The rank-1 definitions reflect a diverse mix of K&K categories, with four Descriptive (A), three Normative (C), two Psychological (D), two Genetic (F), and one Historical (B). Furthermore, while the 60 top positions in Table 5 are occupied by 48 distinct works, specific theoretical clusters recur across related topics. Bidney accounts for the top two slots in *individual acquired behavior* (C-II-3

and C-II-4) and appears again in *material tools and ideas* (A-16). Four definitions produced by the 1941 Committee on Conceptual Integration occupy prominent ranks across *human products and results* (D-I-9, F-I-9, F-I-10) and *cultural order of phenomena* (F-I-11), reflecting their unified operational vocabulary.

4.2 Taxonomy vs. Lexical Reality and K&K's Classification

The decisive empirical test concerns whether K&K's seven deductive categories correspond to the bottom-up thematic structure revealed by LDA. A cross-tabulation of each definition's dominant topic against its K&K category yields a statistically significant association ($\chi^2 = 108.0$, simulated $p = 0.0008$ across 10,000 replications).[8] Yet this departure from pure randomness does not imply that K&K's categories represent cohesive semantic clusters.

The Sankey diagram in Figure 5 maps the flows between K&K's categories (and subgroups) on the left and dominant LDA topics on the right. Rather than a clean, block-diagonal mapping, the diagram reveals pervasive dispersion across theoretical boundaries, visible in three ways.

Read from the category side, the flows show fragmentation within every K&K category. No single category concentrates more than 41% of its definitions within any single topic. Descriptive (A) definitions disperse across nine distinct topics, with only 9 of 22 falling into *customs, beliefs and knowledge*. Normative (C) definitions split across eight topics, with 11 of 27 landing in *way of life of a people*. Psychological (D) definitions, the largest category in the corpus ($N = 39$), disperse widely across ten topics, with at most 10 definitions (26%) congregating in *transmitted group behavior*. Genetic (F) definitions ($N = 43$) scatter across nine topics.

Read from the topic side, the flows reveal the mirror image, heterogeneity within nearly every topic. While *way of life of a people* is heavily Normative (55%) and *customs, beliefs and knowledge* is plurality Descriptive (47%), other major topics like *individual acquired behavior*, *shared group living*, and *sum total of group life* recruit definitions evenly across historical, descriptive, normative, psychological, and genetic categories.

The within-definition mixtures complete the picture. The median definition allocates only 33% of its total probability mass to its primary topic, compared to a baseline uniform distribution of 8.3%. Mid-century definitions are complex, multifaceted mixtures combining several semantic motifs.

The topic distributions lead to an unambiguous conclusion; K&K's deductive taxonomy does not reflect the latent thematic organization of mid-century culture theorizing. While K&K sorted definitions based on perceived philosophical emphasis (e.g., whether a definition highlighted historical transmission or psychological adjustment), practicing theorists routinely integrated multiple thematic motifs within individual definitions. K&K's categories functioned as an external, bibliographic filing system designed to organize an unwieldy monograph, rather than capturing organic boundaries in how social scientists actually defined culture.

5 Discussion

In this paper, I have re-examined Kroeber and Kluckhohn's landmark 1952 corpus through the dual lenses of two-mode network analysis and computational topic modeling. Rather than treating their compilation as a static museum of historical artifacts, approaching the archive through the principle

[8] Expected cell frequencies fall below 5 in 81 of the 84 table cells, precluding standard asymptotic chi-square approximations; the reported significance level is derived via Monte Carlo permutation under the null hypothesis of independence.

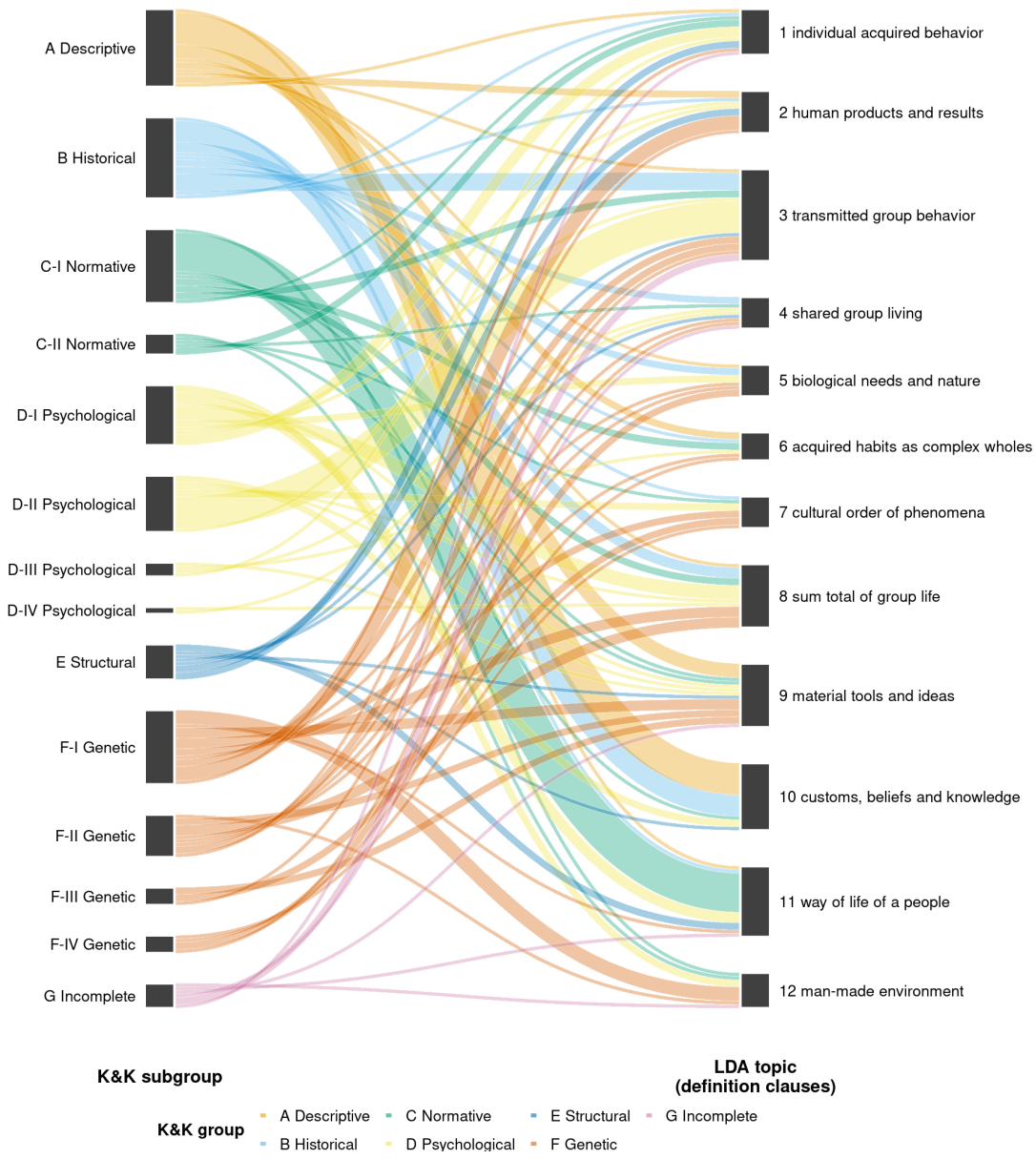


Figure 5: The 171 definitions carrying a text, by their own K&K subgroup (left) and dominant LDA topic (right). Each definition is assigned to the topic with the largest weight. Ribbons take the color of their origin node's category, and node height is the number of definitions. The roman-numeral subgroups are shown within categories C (Normative), D (Psychological) and F (Genetic); categories A (Descriptive), B (Historical), E (Structural) and G (Incomplete) have no subgroup level.

of duality (Breiger, 1974) reveals the latent structural and semantic organization of mid-century culture theorizing. Four broad substantive implications emerge from this investigation.

First, the findings show the analytical value of relational duality for the cognitive history of social scientific concepts. Standard historical accounts of social science typically evaluate concepts through the lens of individual intellectual biographies or broad disciplinary paradigms. By formal-

izing the archive as a bipartite affiliation network, I show how concepts and theoretical statements mutually constitute one another. Concepts achieve institutional traction because they are coordinated across competing definitions; definitions achieve synthetic power because they assemble broad configurations of concepts. The relational approach moves beyond impressionistic textual exegesis, providing a rigorous empirical methodology for mapping the structural evolution of scientific lexicons.

Second, the macro-topology of mid-century culture theorizing reveals a striking core-periphery architecture that resolves a long-standing disciplinary puzzle. Commentators have frequently vacillated between viewing mid-century culture theory as a chaotic babble of incompatible voices or as an incipient consensus. The network structure shows how both impressions could arise simultaneously. At its core, the conceptual space is anchored by a steep, heavy-tailed "ur-vocabulary" centered on collective social life (**social**, **society**, **group**, **behavior**, **ideas**, **customs**). The shared core provided a common currency across rival schools of thought. At the same time, definitions adhered to a disciplined economy of expression, typically combining a small selection of core anchor terms with a handful of specialized peripheral concepts. The culture concept could thus appear unified at its structural core while remaining indefinitely open to specialized variation at its margins.

Third, examining the structural backbone of the network highlights the preeminent importance of *synthetic theorizing*. In network terms, structural centrality does not belong to wordy catalogs or polemical manifestos; it belongs to integrative formulations that span divergent conceptual registers. Gillin (1942) and Bidney (1942, 1946, 1947) emerge as the paramount structural bridges in the corpus because they self-consciously formulated definitions that reconciled psychological and behavioral processes with normative ideals and structural patterns. Bidney's prominent role is particularly instructive. While historical narratives often marginalize Bidney as an idiosyncratic philosophical critic, topologically his holistic formulations served as the primary relational conduits binding the discipline together. Conversely, Tylor's celebrated 1871 definition occupies a peripheral position in the network backbone, illustrating a sharp divergence between ceremonial origin myths and actual structural integration in scientific discourse.

Finally, the computational text analysis delivers a vital cautionary lesson regarding deductive taxonomic schemes. When tested against the verbatim text using latent Dirichlet allocation, K&K's famous seven-category taxonomy fails to recover the natural thematic structure of the definitions. Instead, mid-century definitions were pervasive thematic mixtures, and the actual lexical axes of culture theorizing—transmitted group behavior, shared ways of life, human products, biological needs—cut directly across K&K's categories. K&K's taxonomy was an editorial apparatus designed to sort an extensive bibliographic review, not an organic representation of how practicing social scientists conceptualized culture. Recognizing the disconnect between top-down administrative classifications and bottom-up semantic reality provides a crucial corrective not only for the history of anthropology and sociology, but for contemporary computational analyses of cultural and scientific fields.

Table 5: The five definitions with the largest weight on each LDA topic. Each cell gives the definition's identifier with its weight in parentheses, and below it the work its quotation comes from. Boroch's appendix prints the reprint year 1931 for A-4, and the work is cited here as 1929. The four 1941 definitions (F-I-9, D-I-9, F-I-10 and F-I-11) are all one mimeographed work of the Committee on Conceptual Integration. Topic labels are those of Table 4.

Topic	Label	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
1	individual acquired behavior	C-II-3 (0.51) Bidney 1942	C-II-4 (0.46) Bidney 1946	E-1 (0.39) Willey 1929	B-21 (0.37) Henry 1949	A-10 (0.32) Linton 1936
2	human products and results	D-I-9 (0.74) Committee on Conceptual Integration 1941	F-I-9 (0.69) Committee on Conceptual Integration 1941	F-I-1 (0.60) Groves 1928	F-I-10 (0.56) Committee on Conceptual Integration 1941	D-II-13 (0.45) Hockett 1950
3	transmitted group behavior	D-II-6 (0.67) Benedict 1947	D-II-14 (0.64) Steward 1950	D-II-16 (0.59) Aberle et al. 1950	E-5(b) (0.59) Linton 1945	D-I-5 (0.59) Young 1934
4	shared group living	B-5 (0.36) Bose 1929	G-3 (0.29) Benedict 1934	D-III-4 (0.28) Roberts 1951	E-6 (0.25) Kluckhohn and Kelly 1945	C-II-6 (0.25) Sorokin 1947
5	biological needs and nature	A-11 (0.86) Panunzio 1939	D-I-7 (0.52) Panunzio 1939	B-9 (0.42) Linton 1936	D-I-16 (0.36) Gorer 1949	F-I-16 (0.30) Bidney 1947
6	acquired habits as complex wholes	A-4 (0.44) Benedict 1929	B-8 (0.40) Lowie 1934	A-14 (0.33) Kluckhohn and Kelly 1945	C-I-4 (0.31) Klineberg 1935	A-1 (0.29) Tylor 1871
7	cultural order of phenomena	F-IV-2 (0.67) Ostwald 1915	F-III-4 (0.46) White 1949	F-I-11 (0.46) Committee on Conceptual Integration 1941	F-I-6 (0.38) Warden 1936	D-II-1 (0.37) Wissler 1916
8	sum total of group life	A-10a (0.48) Lowie 1937	F-I-7 (0.41) Sorokin 1937	B-20 (0.38) Kluckhohn 1949	D-I-1 (0.35) Small 1905	C-I-5 (0.35) Firth 1939
9	material tools and ideas	C-I-18 (0.64) Titiev 1949	F-III-2 (0.58) White 1943	A-18 (0.49) Herskovits 1948	C-II-2 (0.47) Thomas 1937	A-16 (0.43) Bidney 1947
10	customs, beliefs and knowledge	A-3(a) (0.64) Dixon 1928	A-2 (0.59) Wissler 1920	A-3(b) (0.58) Dixon 1928	A-12 (0.57) Murray 1943	A-1 (0.52) Tylor 1871
11	way of life of a people	C-I-13 (0.70) Kluckhohn and Leighton 1946	C-I-19 (0.49) Kluckhohn 1951	C-I-5a (0.49) Lynd 1940	C-I-15 (0.47) Lasswell 1948	C-I-9 (0.46) Linton 1945
12	man-made environment	F-I-1a (0.58) Willey 1927	F-I-17 (0.54) Herskovits 1948	F-II-7(b) (0.50) Feibleman 1946	F-I-18 (0.46) Kluckhohn 1949	D-I-15 (0.39) Turney-High 1949

Appendix A Constructing the definition-level data

The analyses rely on a single primary data structure, a table containing one row for each of the 171 K&K definitions. Each row specifies the definition's group and subgroup, its constituent sub-elements, and the citation identifying its quotation. Here I record how I built that table, the methodological choices made along the way, and the empirical basis for each decision. I join two primary sources. K&K's Index A and Index B, extracted from an OCR scan of the 1952 monograph, provide each definition's author, year, and sub-elements. Boroch's (2016) article supplies each definition's quotation along with page pinpoints. Every join follows an explicit rule, and I document the rationale for each judgment in the study's provenance records.

A.1 Sources and the unit

The unit of analysis is the individual definition, corresponding to one printed index entry in K&K and one quotation in Boroch (2016). Boroch's appendix contains 171 definitions. Boroch's Table 2 has 135 columns, 132 of which match an appendix definition. The two sources number definitions differently. Boroch renumbers within each group, so that his A10 (Winston 1933) corresponds to K&K's A-9. I therefore match definitions by author, year, group, and subgroup, retaining positional agreement strictly as a diagnostic check. Of the 100 matches established by author and year, 48 agree with the positional ordering.

A.2 Index A and Index B

I extracted Index A (authors) and Index B (sub-elements) from the OCR text layer and corrected them using explicit rules. I document these corrections in a manual-fix table of 142 rows, each verified against printed-page evidence. A further 10 rows recover definitions that OCR processing merged into element names, and 14 rows record author-name disambiguations. I avoid general character-substitution rules, because heuristic replacements (such as mapping 'y' to '5') silently generate spurious definitions from unrelated tokens. Each manual fix is keyed to a parsed token, and automated checks confirm that every key matches an existing token.

The target universe comprises 171 definitions. Of these, 169 appear in K&K's author index. The remaining two, C-I-18a (Maquet 1949) and D-II-16 (Aberle et al. 1950), are cited in main-text footnotes but omitted from the author index; I recover their Index B affiliations directly from the sub-element listings that reference them. Four Part III statement definitions (C-5, D-5, D-7, and E-17) are excluded, because they represent K&K's own analytical statements rather than Part II definitions.

A.3 Matching Boroch's definitions to K&K's

The crosswalk matches each column of Boroch's Table 2 to a K&K definition. Of the 135 columns, 117 match automatically when exactly one definition satisfies the criteria, and 15 more are resolved through manual inspection, each documented with its empirical rationale. Three of Boroch's Table 2 columns have no appendix entry of their own and carry no K&K definition, namely A10a, C1-5a, and F4-6. Each is a verbatim duplicate of another quotation, leaving 132 columns with an authentic K&K definition. Specifically, A10a duplicates A10 (Winston 1933), C1-5a duplicates C1-5 (Firth 1939), and F4-6 duplicates F4-5 (an entry that Boroch's numbering mislabels as F4-6). Because none of these three adds an independent quotation, I exclude them from the network. The F4-5 quotation corresponds to Kluckhohn and Kelly 1945 (printed p. 87), yielding K&K's F-IV-5.

Table 6: The four split definitions and their variants.

K&K definition	Variant	Boroch definition and source	Page numbers
A-3	A-3(a)	A3, Dixon 1928	3
A-3	A-3(b)	A4, Dixon 1928	3
E-5	E-5(a)	E5, Linton 1945 (footnote 5)	5
E-5	E-5(b)	E6, Linton 1945	32
F-II-4	F-II-4(a)	F2-4, Blumenthal 1937 (footnote 4)	3
F-II-4	F-II-4(b)	F2-5, Blumenthal 1937	12
F-II-7	F-II-7(a)	F2-8, Feibleman 1946 (tentative; footnote 7)	75
F-II-7	F-II-7(b)	F2-9, Feibleman 1946 (final)	76

K&K's printed definitions A-10a (Lowie 1937) and C-I-5a (Lynd 1940) draw their quotations from appendix entries A12 and C1-6.

A.4 Quotations and citation keys

Each analyzed definition is linked to a single quotation. Boroch supplies 170 of these quotations, comprising the 132 definitions from Boroch's Table 2 and the 39 appendix-only definitions, less the Sears addendum (C1-22), which attaches to a specialized commentary rather than a core definition. Definition D-III-4 lacks a Boroch entry; its quotation is taken from K&K's printed footnote 14 (Roberts 1951, pp. 3 and 6, printed p. 60). Four quotations are not in English and are rendered from a translation table, with B-17 and C-I-18a translated from French, and F-I-5 and F-II-3 from German. C-I-18a adopts K&K's own printed English translation, while the remaining three are working translations awaiting page-image verification.

Each definition is uniquely identified by a composite citation key combining Boroch's author, year, and page pinpoints with the opening six words of the quotation. The composite key is complete and unique across all 171 definitions. The page numbers reflect Boroch's printed citations and have not been independently verified against the primary volumes; the key therefore identifies each quotation as recorded in Boroch's compilation.

A.5 Splitting the composite quotations

Boroch quotes parts of four K&K definitions separately, designating them as (a) and (b). Because K&K provide only a single index entry for each of these composite entries, I split each of the four definitions into two distinct variants (Table 6). The division follows Boroch's textual partitions rather than K&K's indexing. Each variant receives its own quotation and unique citation key.

Because K&K provide a single index entry per composite definition, each variant initially inherits a duplicate of its parent's sub-element links, totaling 41 ties. I then apply a lexical filter to assign 29 of these 41 ties to a specific variant. A link is retained on a variant if its quotation contains the sub-element's textual term and omitted from the other. The 12 links where both or neither quotation contains the term are retained across both variants. Because this filter relies on exact word matching, generic terms (such as "total" or "past") may occasionally be misallocated. Each assignment represents an interpretive reading of the quotation text. Pruning these redundant ties also removes the corresponding heading affiliations, accounting for 34 edge rows. In total, splitting the four composite entries expands the verified network from 167 to 171 definitions and from 1189 to 1201 edges.

A.6 Checks, limitations, and open items

The reconstructed networks depend on the transcription of K&K's Index B and the verified-definition filtering criteria. Because element names are not unique across K&K's taxonomic headings, identical sub-element labels are merged, which may occasionally link conceptually distinct uses. Several readings rely on OCR corrections recorded in the project's manual-fix tables. Two definitions omitted from Index A are included, namely C-I-18a and D-II-16, identified from main-text discussions. Conversely, K&K's Part III statement codes (C-5, D-5, D-7, and E-17) are excluded, as they represent authorial commentary rather than definitions. The four split definitions and the 29 variant-assignment decisions represent interpretive judgments (Appendix Appendix A); because the variants share a single printed index entry in K&K, assigning concept ties based on quotation text is an analytical intervention that readers can evaluate directly against the quoted passages. The definition-level spanning trees and topic models reflect these structural decisions. Because projection tie weights frequently recur, maximum spanning trees are not unique. Betweenness centrality is therefore averaged across an ensemble of 1000 random trees; the trees illustrated in the figures depict the single representative member closest to the ensemble mean, such that local neighborhood topology reflects that specific realization, while ranks below the leading positions in Table 3 exhibit sensitivity to tie-breaking.

Automated validation routines verify network topology, round-trip consistency, and attribute integrity, though they do not independently evaluate the substantive content of individual edges. Block validation places 1301 of the 1326 Index B edge rows (prior to the 34-row variant pruning) in their expected printed block, with 25 placed in adjacent blocks; page-image inspection confirmed 23 of those 25. Several items remain open for future archival reconciliation: verifying the publication year of footnote 5 (attributed by Boroch to Linton 1945a) against high-resolution scans, resolving duplicate columns in the crosswalk, verifying the three non-English translations against primary editions, and confirming Boroch's page pinpoints directly against original source texts.

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